

五 19, 22.25

19.

$$S_X(\omega) = \frac{1}{(\omega^2 + 2)(\omega^2 + 3)}$$

$$= \frac{\frac{\sqrt{2}}{4}}{\omega^2 + 2} - \frac{\frac{2\sqrt{3}}{8}}{\omega^2 + 3} = \frac{\sqrt{2}}{4(\omega^2 + 2)} - \frac{\sqrt{3}}{4(\omega^2 + 3)}$$

$$R_X(\tau) = \frac{\sqrt{2}}{4} e^{-\sqrt{2}|\tau|} - \frac{\sqrt{3}}{8} e^{-\sqrt{3}|\tau|}$$

> 5.

$$R_Y(\tau) = E[X(t)X(t+\tau)] \quad \mu_X(t) = \mu_Y(t) = 0 = \mu_Y$$

$$R_Y(\tau, t+\tau) = E[(X(t+L)-X(t))X(t+L+\tau-X(t+\tau))]$$

$$= 2R_X(\tau) - R_X(\tau-L) - R_X(\tau+L)$$

$$R_X(\tau) \xrightarrow{F} S_X(\omega)$$

$$R_X(\tau-L) \xrightarrow{F} S_X(\omega) e^{-i\omega L}$$

$$R_X(\tau+L) \xrightarrow{F} S_X(\omega) e^{i\omega L}$$

$$R_Y(t, t+\tau) = R_Y(\tau) \quad \tau \text{ 平稳过程}$$

$$S_Y(\omega) = F(R_Y(\tau))$$

$$R_Y(\tau) = 2S_X(\omega) - S_X(\omega) \cdot 2\cos\omega L$$

$$= 2S_X(\omega)(1 - \cos\omega L)$$

22.

$$1 \xrightarrow{F} 2\pi\delta(\omega)$$

$$\frac{1}{\pi} \xrightarrow{F} 2\delta(\omega)$$

$$R_X(t) = \frac{1}{\pi} + \frac{1}{2\pi} \int_{-1}^1 (1-w)e^{i\omega t} d\omega$$

$$= \frac{1}{\pi} + \frac{1}{2\pi} \left[2 - \left(\int_0^1 w e^{i\omega t} d\omega - \int_{-1}^0 w e^{i\omega t} d\omega \right) \right]$$

$$= \frac{1}{\pi} + \frac{1}{2\pi} \left[2 - 2 \left(\frac{\sin t}{t} + \frac{\cos t - 1}{t^2} \right) \right]$$

$$= \frac{2}{\pi} + \frac{1}{\pi} \left(\frac{\sin t}{t} + \frac{\cos t - 1}{t^2} \right)$$

$$R_X(t) = \frac{2}{\pi} - \frac{1}{\pi} \left(\frac{\sin t}{t} + \frac{\cos t - 1}{t^2} \right)$$

$$\frac{\Delta}{\Delta t} = \sin \Delta$$

$$\lim_{\Delta \rightarrow 0} \frac{\Delta}{\Delta t} = 1 = \sin 0$$

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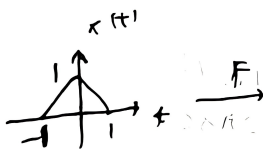
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22.



$$\frac{1}{2\pi} S_X(\omega) \xrightarrow{F} \frac{1}{\pi} S_X(\omega)$$

$$R_X(\tau) = \frac{1}{2\pi} S_X(\omega) + \frac{1}{\pi}$$

$$= \frac{1}{\pi} + \frac{4}{2\pi} \frac{\sin^2 \frac{\tau}{2}}{\tau^2} = \frac{1}{\pi} + \frac{2}{\pi} \frac{\sin^2 \frac{\tau}{2}}{\tau^2}$$

$$= \frac{1}{\pi} \left(1 + 2 \frac{\sin^2 \frac{\tau}{2}}{\tau^2} \right)$$